

Four Colour Theorem (4CT):

Any planar, simple connected graph can be coloured by at most 4 colours.

- First computer assisted proof in mathematics.
- Won't be going through all 633 cases.

Conjectured in 1852, popularized by De Morgan.

Kempe created a false proof in 1879, disproved in 1890 by Heawood.

- Maximal planar graph. A maximal planar graph will subject the graph to the most colouring restrictions.
 - Every planar graph is contained in maximal planar graph.
- Euler formula:
 - # of regions = # edges - # vertices + 2
- Euler's formula for a planar graph: $E \leq 3V - 6$

Textbook: $2e = \left[\sum_{r_i} \deg(r_i) \right] \geq 3r$ for connected, planar, simple graphs. Each region has at least 3 edges.

$\Rightarrow \frac{2}{3}e \geq r$

$\Rightarrow \frac{2}{3}e \geq e - v + 2$

$\Rightarrow e \leq 3v - 6$. K_5 is non-planar. $e = 10 \neq 3v - 6 = 9$

$K_{3,3}$ satisfies degree at region must be at least 4. $\downarrow$ * Proof this is false

If a connected, planar, simple graph has e edges & v vertices $v \geq 3$ and no circles of length 3, $e \leq 2v - 4$, provable since $2e = \sum_{r_i} \deg(r_i) \geq 4r$

$v \geq 3$,

1 region: tree, no forest. $E = V - 1$

Note:

- Think of sphere (sphere is bounded) = plane.
 - Stereographic projection.
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$E = 3V - 6$ for a maximally planar graph.

- Each region has 3 edges from above.

We want the average degree per node, or arity (in a map, not dual graph),

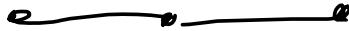
$$2E = 6V - 12 = \text{Sum of degrees}$$

Therefore, average degree is $(6 - 12/V) < 6$

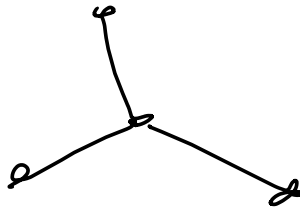
- I think every maximally planar graph has a vertex of degree < 6 .

Kempe's proof by induction:

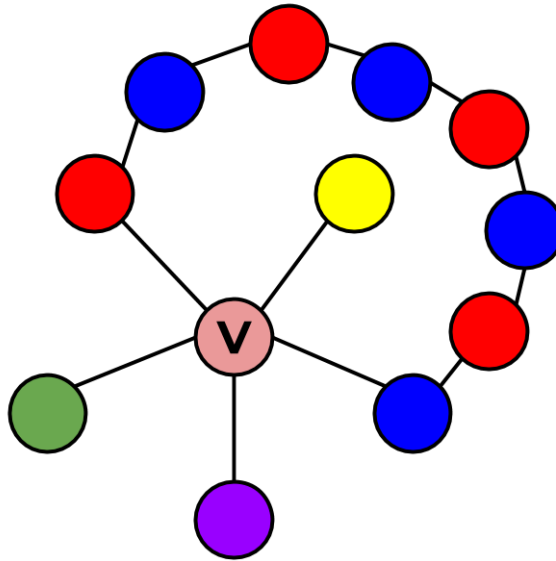
Degree = 2:



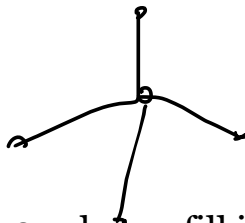
Degree = 3:



Kempe Chain:



Degree = 4:



If ring is coloured with 3 colours, fill in centre node with remaining colour.

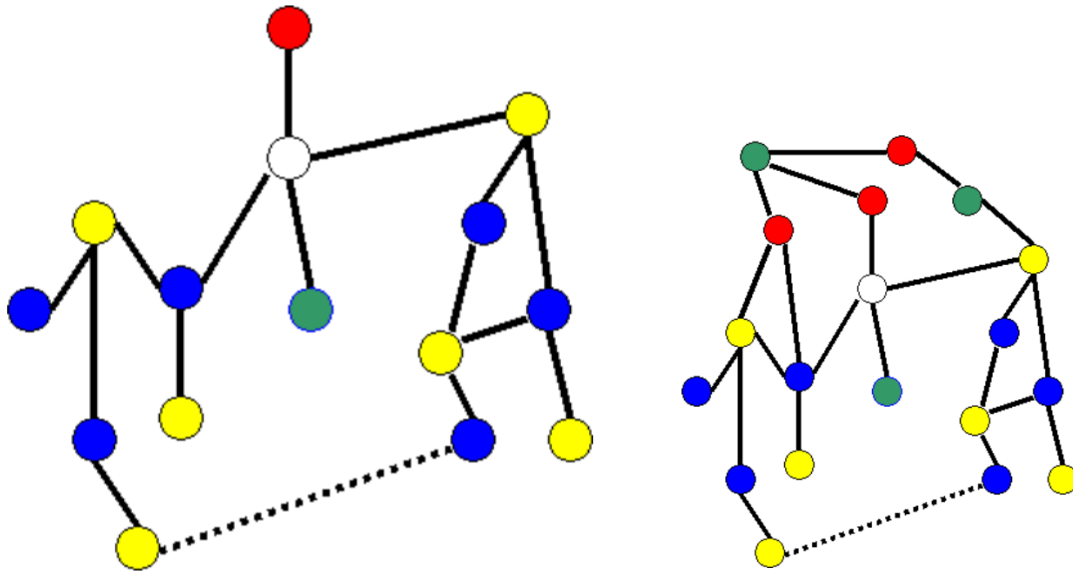
Kempe's "Proof" by Induction of the Four Color Theorem

Kempe's proof is constructive in the sense that the details of the proof imply a recursive algorithm for coloring. Kempe's proof is by induction on the number of regions in the map (or nodes in the graph). Like many difficult inductive proofs, he uses *strong* induction. This means he attempts to show that a graph with n nodes can be four-colored and assumes that all graphs with fewer than n nodes can be four-colored. This style is in mild contrast to the simple inductive proofs that we use for formulae like: $1 + 2 + 3 + \dots + n = n(n+1)/2$ where we start with n th case and work toward the $n+1$ st case.

Before he gets to the inductive idea, Kempe begins with a clever way to categorize the possible graphs. He first states that we might as well consider graphs all of whose faces are triangles. These *maximal planar* graphs include the maximum number of edges possible, and if they can be four-colored, then so can other graphs with fewer edges. He then proves a lemma (a helper theorem) that any maximal planar graph must have at least one node with degree 5 or less. The proof uses Euler's Theorem for planar graphs and a counting argument to show that if all the nodes had degree 6 or higher, then the graph would not be planar. The details can be found below but they are not relevant to the main thread of the proof, and can be read independently.

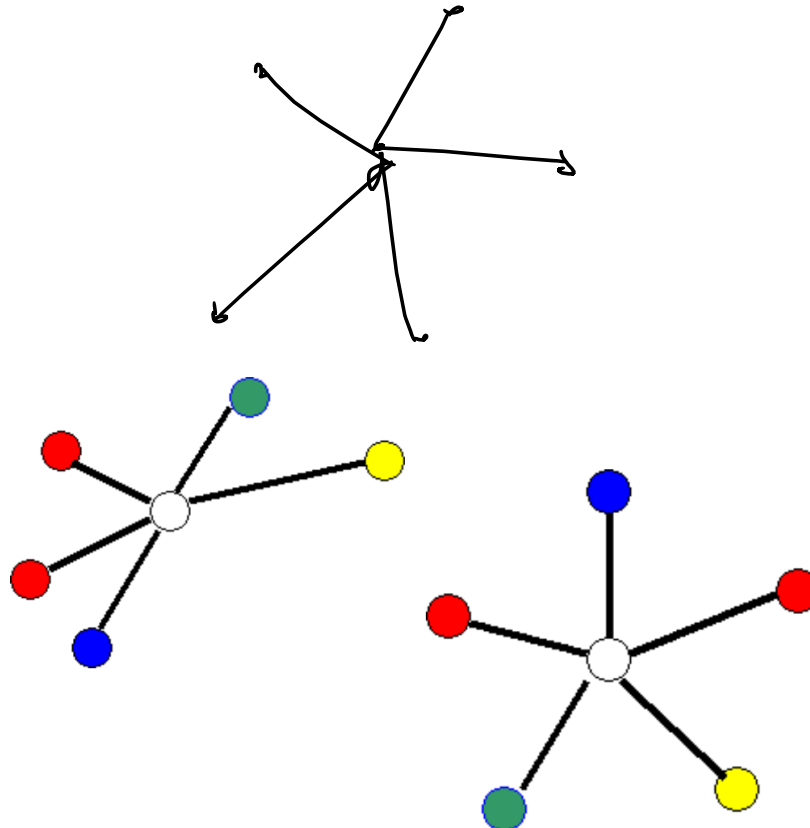
If we denote the number of nodes, edges, and faces (i.e., the bounded regions) of a planar graph by V , E , and F respectively, then Euler's theorem for a plane (or a sphere) is $V - E + F = 2$ (the outside face counts too). Each face of a maximal planar graph is bounded by three edges, and each edge is on the boundary of two faces, so we have $F = 2E/3$. Euler's formula for a complete planar graph then becomes simply $E = 3V - 6$. The *degree* of a node is the number of edges incident to it. Now each edge is connected to two nodes, so the sum of the degrees of all the nodes $2E = 6V - 12$, and hence the average degree per node is $6 - 12/V < 6$, (assuming the graph is finite). This implies that at least one vertex has degree five or less.

See into the induction.

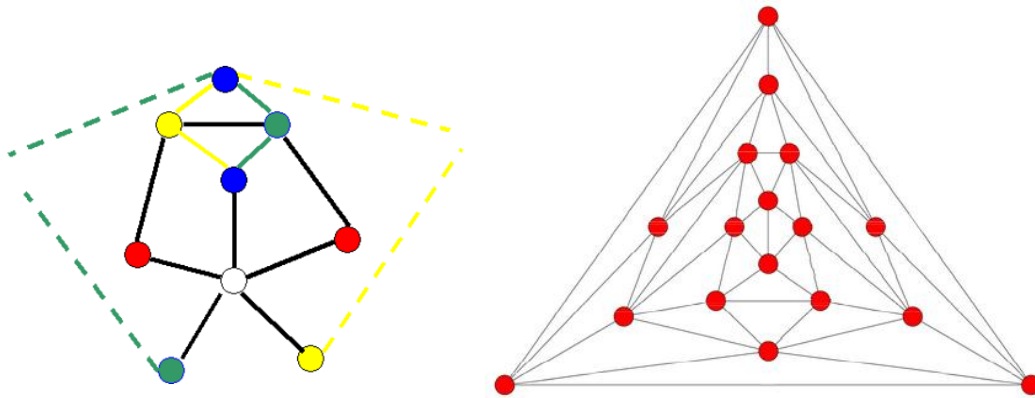


We don't look at adjacent nodes because it is a worst case.

Degree = 5:



Disproof:



Disproof model on left, Errera graph on right (full disproof model).

<https://sites.google.com/site/shaisimonson/professional-page/mathexp/lab4>

Minimal counterexample: 6 connected triangulations

Irreducibility:

Unavoidability: A set of configurations that must exist

A triangulation: Maximally planar graph

Next:

Ask about normal subgroups from SYDE 710

Understand the induction more

Irreducibility and unavoidability

AMS and Robertson's paper

Minimal counterexample (2.1) in Robertson